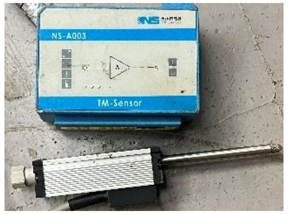
<table>
<tr><th>Style</th><th colspan="2">Type</th><th>Range</th><th colspan="2">Photo</th></tr>
<tr><td rowspan="6">Linear potentiometer (LP)</td><td rowspan="4">Novotechnik</td><td>TEX-0100-415 002-101</td><td>100mm</td><td></td><td></td></tr>
<tr><td>TEX-0150-415 002-101</td><td>150mm</td><td></td><td></td></tr>
<tr><td>TEX-0200-415 002-101</td><td>200mm</td><td></td><td></td></tr>
<tr><td>TEX-0250-415 002-101</td><td>250mm</td><td></td><td></td></tr>
<tr><td rowspan="2">TM Automation Instruments</td><td>NS-WY02</td><td>25mm</td><td></td><td>
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<tr><td>NS-WY02</td><td>100mm</td><td></td><td>
</td></tr>
<tr><td rowspan="3">String potentiometer (SP)</td><td rowspan="3">TM Automation Instruments</td><td>NS-WY06</td><td>200mm</td><td></td><td></td></tr>
<tr><td>NS-WY06</td><td>500mm</td><td></td><td></td></tr>
<tr><td>NS-WY06</td><td>1000mm</td><td></td><td></td></tr>
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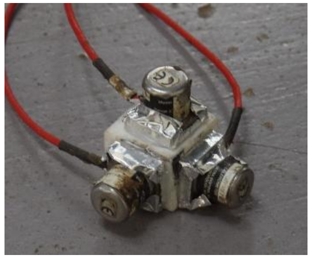
Accelerometers	Setra Systems	141	8g		
Load cells		BHR-4B	20t		
Load pins					
Strain gauges					

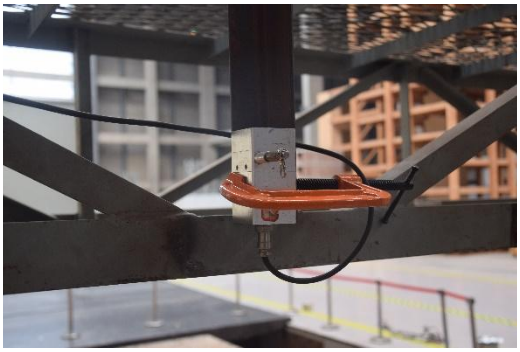
1. Global response

1.1. Accelerometer

The accelerometers were mounted in sets of three to measure the NS, EW and vertical directions. The three accelerometers were glued on three surfaces of a 30×30×30 mm cube and then glued to an angle steel that bolted to the wall panel.

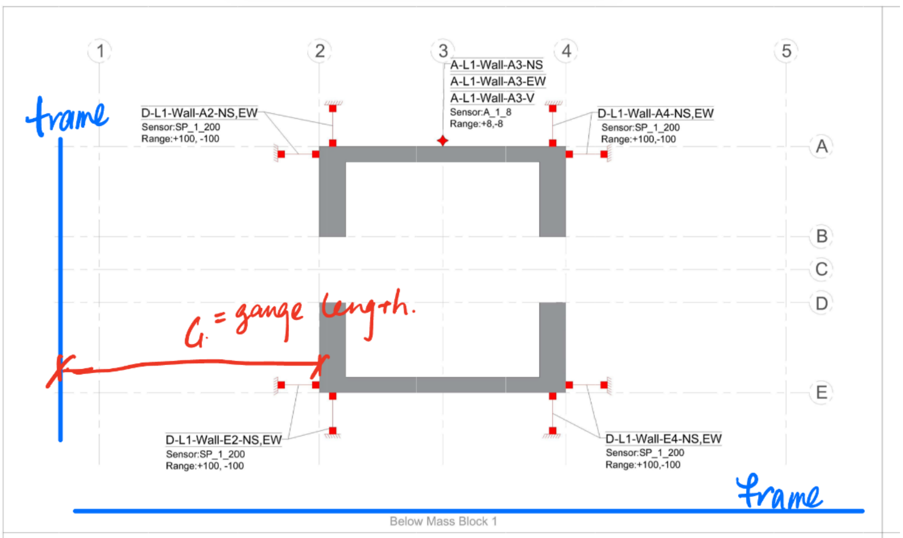
1.2. Displacement

String pots were mounted from a reference frame erected around the building (off the table) with string/wire connected by a screw into the inter-layer wall top corners.

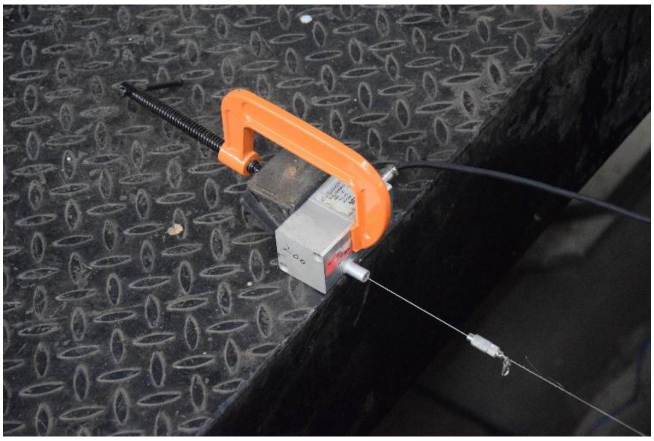


String pot mounted on reference frame

Screws at wall top corners



Also, a string pot was mounted at the edge of shake table to monitor the table displacement.



G = gauge length (including wires)

2. Unintended response

2.1. Foundation displacement

Linear pots with spring were mounted at the foundation corner in three directions (NS, EW and vertical).

2.2. Outrigger beam uplift

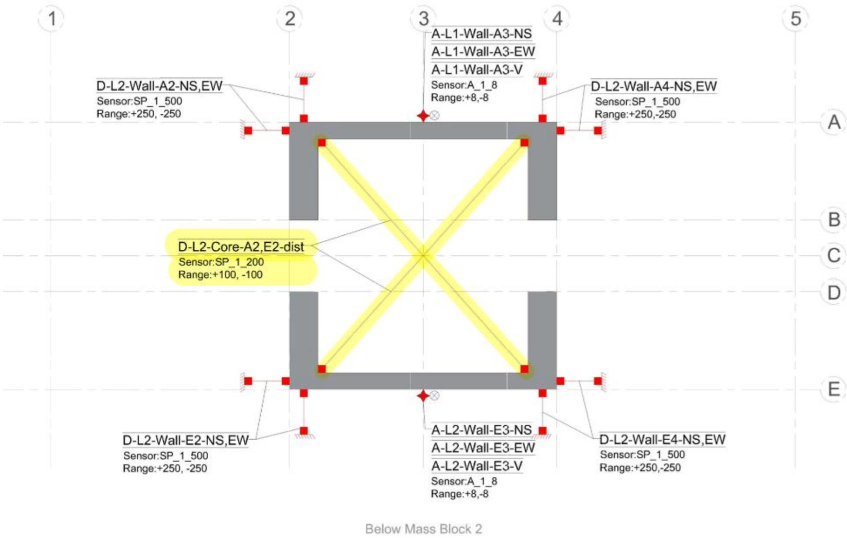
Linear pots with spring were mounted on the wall top connecting the wall and outrigger beam.

2.3. Core welding uplift

Linear pots with spring were mounted connecting the two panels at the welding seam.

2.4. Core distortion

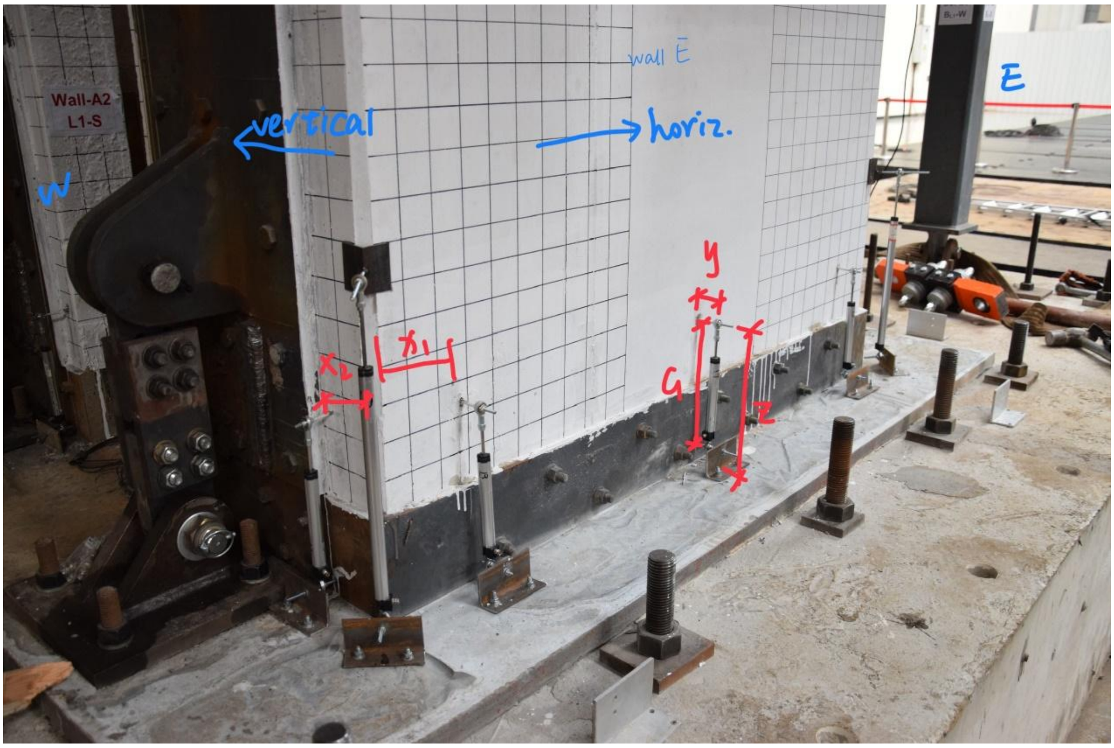
String pots were mounted at the web-to-flange corner in the core with string/wire, connected with a screw into the wall panel. Sensors were installed at the 2, 4, 6 and 8 levels.

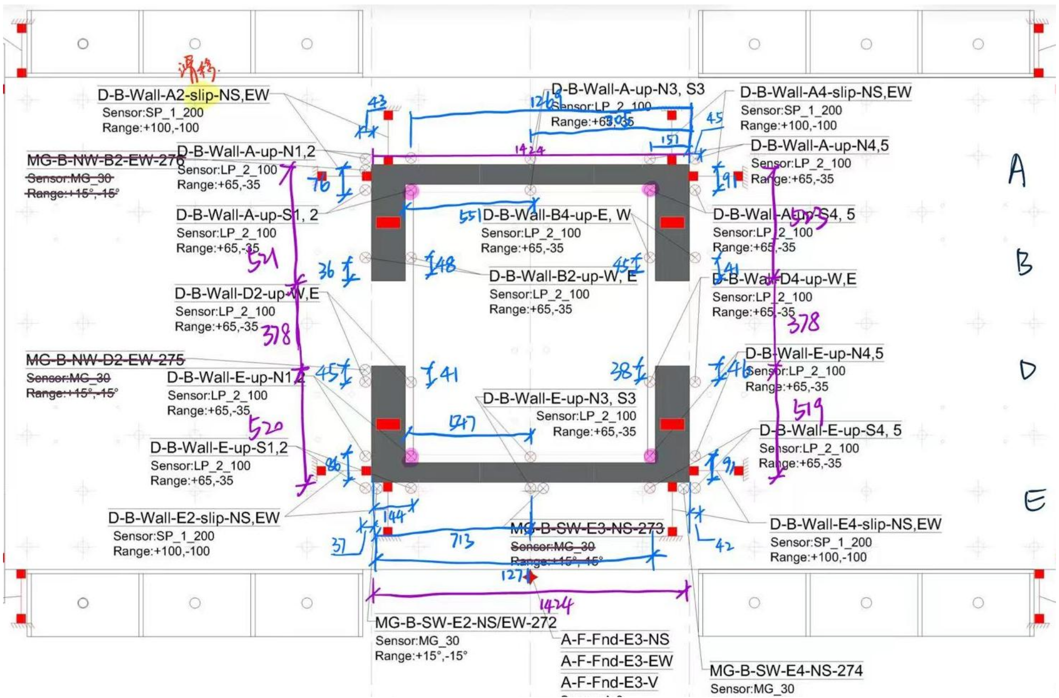


3. Local response

3.1. Wall uplift

Linear pots with rod ends were mounted at the top end on the threaded rods epoxy-glued into the wall (about 30mm). Bottom ends mounted to threaded rod secured to aluminum angle mounted on the foundation with expansion screws.





member1 = wall length

member2 = wall thickness

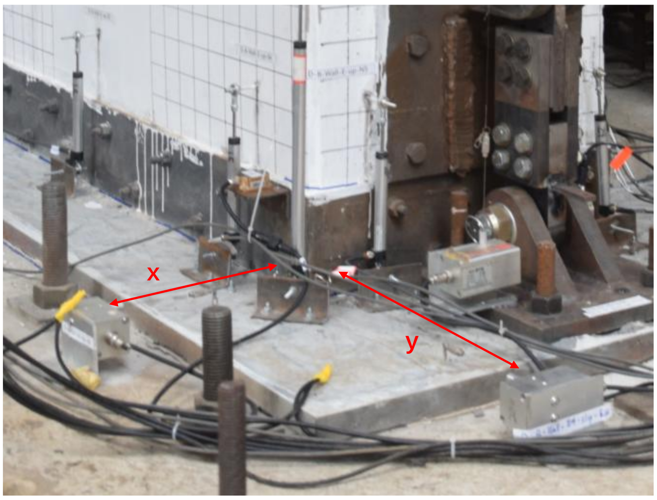
x1 = horizontal distance from wall edge (E or W end – as per numbering of LP) to threaded rod in wall

x2 = vertical dist. from wall edge (N or S end) to threaded rod in wall

y = offset of LP rod from face of wall
z = height of wall threaded rod measured from the bottom edge of the wall
G = gauge length (dist. between bearings)

3.2. Wall slip

String pots were mounted on the foundation with angle steel connecting the wall corner by wire.



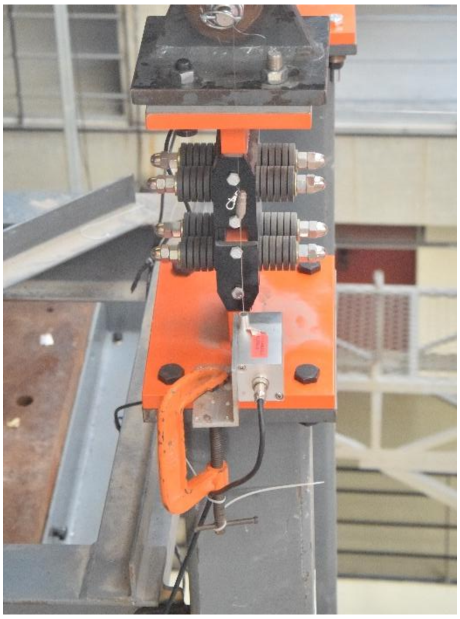
G = gauge length (distance between wall face and sensor)

3.3. PT tendon force

Load cells were mounted between the channel steel beam and anchorage with two 12mm thick steel plate clamped.

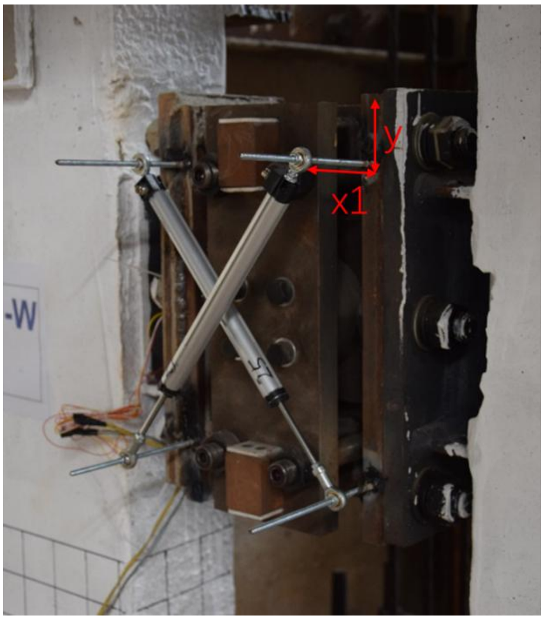
3.4. Damper (W2F & B2C) displacement

String pots were attached to the dampers by angle aluminum glued to the bottom plate and wire. However, the glue was not strong enough and fell off during the test. With original one re-glued, another string pot was then added by clamping to the damper.

3.5. Coupling beam displacement

Linear pots with end rods were attached to the coupling beam in diagonal directions. The screw rods were welded to the coupling beam, with the same distance to the end of SCFD.



x1 = offset from damper to center of end rod

y = offset from damper edge

3.6. Coupling beam strain

Strain gauges at 0 and 45 degrees were attached to the embedded connecting plate.

3.7. Outrigger beam strain

Strain gauges were glued to the top face of outrigger beam in the longitudinal direction.